

Short Question

1. What is computer graphics? Explain different application areas of computer graphics.
2. How can you draw a circle using mid-point circle algorithm? Explain with the algorithm.
3. Explain 3D basic geometric transformation with an example.
4. What is polygon clipping? Explain Sutherland Hodgman algorithm for polygon clipping.
5. Given a triangle with vertices A(2,3), B(5,5), C(4,3) by rotating 90 degree about the origin and then translating two unit in each direction. Use homogenous transformation matrix to find the new vertices of the triangle Note.
6. Define horizontal and vertical retrace of electron beam. Explain architecture of random scan display system.
7. How does decision parameter can be used to draw circle? Calculate the points to draw a circle having radius 5 and center (10,5) using midpoint circle algorithm.
8. What do you mean by window to viewport transformation? Explain 2D viewing pipeline.
9. Generate the points between the end points of a line (2,2) and (9,6) using DDA line drawing algorithm.
10. Apply Liang Barsky Line Clipping algorithm to the line with coordinates (30, 60) and (60, 25) against the window $(x_{wmin}, y_{wmin}) = (10, 10)$ and $(x_{wmax}, y_{wmax}) = (50, 50)$.
11. Use the Cohen-Sutherland algorithm to clip the line P1(70, 20) and P2(100, 10) against a window lower left hand corner (50, 10) and upper right hand corner (80, 40).
12. A homogenous coordinate point P(3, 2, 1) is translated in x, y, z direction by -2, -2 & -2 unit respectively followed by successive rotation of 60° about x- axis. Find the final position of homogenous coordinate.

Long Question

1. Explain the working details of DDA line drawing algorithm? Digitize the line with endpoints (2,2) and (10,5) using Bresenham's line drawing algorithm.
2. Derive the formula for windows to viewport transformation. Given a window bordered by (0,0) at the lower-left and (4,4) at the upper right. Similarly, a viewport bordered by (0,0) at the lower-left and (2,2) at the upper right. If a window at position (2,4) is mapped into the viewport.
3. What will be the position of viewport to maintain same relative placement as in window?
4. What is the need of homogeneous coordinate system in geometric transformation system? Find the new coordinates of rectangle ABCD whose center is at (4,2), reduced to half of its size while the center remains same. Coordinates: A(0,0), B(0,4), C(8,4), D(8,0).
5. Explain Sutherland Hodgeman Polygon Clipping algorithm and trace it to clip a polygon with end points P(10,30), Q(25,35), R(35,50), S(60,30), T(45,28), U(40,15) against a window whose lower-left corner is (15,25) and upper-right corner is (50,40).

Assignment

Dinesh Poudel

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1) Computer graphics is the process of drawing pictures and creating images with the help of programming. It is used to create, manipulate and represent the graphics and visual images. It also involves the generation of picture and animation using specialized hardware and software.

The different application areas of CG are:

i) Entertainment Industry: It is used to create animated movies, videos and games

ii) Education and Training: Used in e-learning, simulations and virtual classrooms to enhance understanding.

iii) Medical Field: Used by doctors for finding the diseases using CT-scan, MRI and surgical simulation.

iv) GUI: It is used in OS and software application to make them user friendly.

v) Business & advertisement: Used for product design, digital marketing and presentation

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2) The midpoint circle algorithm is used to draw a circle effectively and efficiently. It determines the points needed to draw the circle using decision parameter.

Algorithm:

1) Start

2) Input radius r , centre (x_c, y_c)

3) Initialize: $x = 0$, $y = r$, $p = 1 - r$

4) Plot initial point (x, y)

5) Repeat until $(x \geq y)$:

• increment: $x = x + 1$

• if $p < 0$

$p = p + 2x + 1$

else

$y = y - 1$

$p = p + 2x + 1 - 2y$

plot points using symmetry

$(x_c \pm x, y_c \pm y), (x_c \pm y, y_c \pm x)$

6) Stop when $(x \geq y)$

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3) In computer graphics, 3D geometric transformations are used to manipulate objects in three-dimensional space (x, y, z) .

Types of basic 3D transformation:

1) Translation: It is the process of moving an object from one point to another without changing its shape.

$$x' = x + tx, \quad y' = y + ty, \quad z' = z + tz$$

example:

if point $P(2, 2, 3)$ is translated by $(4, 5, 6)$

$$x' = 2 + 4 = 6, \quad y' = 2 + 5 = 7, \quad z' = 3 + 6 = 9$$

$$P' = (6, 7, 9)$$

2) Scaling: It is the process of changing the size of an object.

$$x' = x \cdot S_x, \quad y' = y \cdot S_y, \quad z' = z \cdot S_z$$

example:

if $P(2, 2, 3)$ is scaled by $(2, 2, 2)$

$$P' = (4, 4, 6)$$

3) Rotation: It is the process of turning an object around (x, y, z) axis. Rotation about Z-axis.

$$x' = x \cos \theta - y \sin \theta$$

$$y' = x \sin \theta + y \cos \theta$$

$$z' = z$$

example:

Rotate point $P(1, 0, 0)$ by 90° about z -axis,

$$P' = (0, 1, 0)$$

4) Reflection: It produces mirror image of object

About xy -plane: $(x, y, -z)$

About yz -plane: $(-x, y, z)$

About xz -plane: $(x, -y, z)$

example:

Reflect $P(2, 3, 4)$ about xy -plane

$$P' = (2, 3, -4)$$

5) Shearing: It means slanting or skewing an object in particular direction.

x -shear

$$x' = x + Sh_x y + Sh_z z$$

Homogeneous coordinates:

$$(x, y, z, 1)$$

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4) Polygon clipping is the process of removing the parts of the polygon that lie outside a clipping window and keeping only the visible portion inside window.

The Sutherland - Hodgman algorithm is used to clip a polygon against a rectangular window.

Algorithm:

- 1) Input the polygon vertices and clipping window bound axes (x_{min} , x_{max} , y_{min} , y_{max})
- 2) Select a clipping boundary (L, R, B, T)
- 3) Take each edge of the polygon one by one (i.e. From vertex P_1 to P_2)
- 4) Check the position of vertices with respect to boundary
 - If both P_1 and P_2 are inside $\rightarrow$ output P_2
 - If P_1 inside & P_2 outside $\rightarrow$ output intersection point
 - If P_1 outside & P_2 inside $\rightarrow$ output intersection point & P_2
 - If both outside $\rightarrow$ output nothing
- 5) Store the output vertices after processing all edges
- 6) Repeat steps 2-5 for all four boundaries
- 7) The final set of vertices obtained after all stages is the clipped polygon.

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5) Soln:

$$\text{Object of } \triangle ABC = \begin{pmatrix} 2 & 5 & 4 \\ 3 & 5 & 3 \\ 0 & 0 & 1 \end{pmatrix}$$

Composite transformation matrix = $T(2,2) \times R(90)$

$$= \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix} \times \begin{pmatrix} \cos 90^\circ & -\sin 90^\circ & 0 \\ \sin 90^\circ & \cos 90^\circ & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \times \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

New coordinate:

$$= \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \times \begin{pmatrix} 2 & 5 & 4 \\ 3 & 5 & 3 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} -3 & -5 & -4 \\ 2 & 5 & 3 \\ 0 & 0 & 1 \end{pmatrix}$$



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- 6) Horizontal and vertical retrace of an electron beam refer to the return movement of the beam in a display system after completing a scan. In horizontal retrace, the electron beam moves from the right end of a scan line back to the left side to begin drawing the next line. During this time the beam is turned off to avoid drawing unwanted lines. In vertical retrace the beam returns from the bottom-right corner of screen to the top-left corner after completing one full frame.

The random scan (vector) display system draws images by directly drawing lines between points instead of scanning the whole screen. It consists of a display processor, refresh, buffer, CRT. The processor reads line data, DAC convert it to signals and the electron beam draws lines on the screen. It produces smooth, high-quality images but is not suitable for complex filled graphics.

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7) The decision parameter (p) is used to decide which pixel is closer to the actual circle path at each step. At every step we choose between:

Soln

$$\text{Center} = 10, 5$$

$$\text{radius } (r) = 5$$

$$p = 1 - r = 1 - 5 = -4$$

$$\text{Initial point } (0, r) = (0, 5)$$

Here,

$$p_k > 0, \quad p_{k+1} = p_k + 2x_{k+1} + 1 - 2y_{k+1} \quad (x_{k+1}, y_{k+1})$$

$$p_k < 0, \quad p_{k+1} = p_k + 2x_{k+1} \quad (x_{k+1}, y)$$

K	p_k	x_{k+1}	y_{k+1}	$2x_{k+1}$	$2y_{k+1}$
0	-4	1	5	2	10
1	-1	2	5	4	10
2	4	3	4	6	8
3	3	4	3	8	6

- 8) A window in computer graphics is a selected rectangular area in a world coordinate system that defines what ~~portion~~ portion of the scene is to be display whereas a viewport is the rectangular area on the display screen where the contents of the windows are mapped and shown.

The 2D viewing transformation pipeline is a sequence of steps used in computer graphics to convert objects from the world coordinate system to screen coordinates so that they can be displayed on a device. Steps in pipeline flow:

world coordinates $\rightarrow$ window $\rightarrow$ clipping $\rightarrow$
Normalization $\rightarrow$ viewport $\rightarrow$ screen display

- 9) DDA line drawing algorithm:

Given,

$$(x_1, y_1) = 2, 2$$

$$(x_2, y_2) = 9, 6$$

$$dx = x_2 - x_1 = 9 - 2 = 7$$

$$dy = y_2 - y_1 = 6 - 2 = 4$$

$$\text{max steps} = \max(dx, dy) \\ = 7$$

$$x_{inc} = dx / \text{steps} = 7 / 7 = 1$$

$$y_{inc} = dy / \text{steps} = 4 / 7 = 0.571$$

$$x_1 = x_0 + x_{inc}, \quad y_1 = y_0 + y_{inc}$$

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x	y	Plot (x,y)
3	2.571	(3, 2.57)
4	3.142	(4, 3.142)
5	3.713	(5, 3.713)
6	4.284	(6, 4.284)
7	4.855	(7, 4.855)
8	5.426	(8, 5.426)
9	6	(9, 6)

10) Soln:

$$x_{wmin} = 10, \quad x_{wmax} = 50$$

$$y_{wmin} = 10, \quad y_{wmax} = 50$$

Point P(30, 60)

$$x - x_{wmin} = 30 - 10 = 0 \quad L$$

$$x_{wmax} - x = 50 - 30 = 0 \quad R$$

$$y - y_{wmin} = 60 - 10 = 0 \quad B$$

$$y_{wmax} - y = 50 - 60 = -1 \quad T$$

Point Q(60, 25)

$$x - x_{wmin} = 60 - 10 = 0$$

$$x_{wmax} - x = 50 - 60 = -1$$

$$y - y_{wmin} = 25 - 10 = 0$$

$$y_{wmax} - y = 50 - 25 = 1$$

10) Soln;



Here,

$$x_1 = 30$$

$$y_1 = 60$$

$$x_2 = 60$$

$$y_2 = 25$$

$$x_{wmin} = 20$$

$$y_{wmin} = 20$$

$$x_{wmax} = 50$$

$$y_{wmax} = 50$$

then,

first we calculate P_k and Q_k where,

$$P_1 = \Delta x = x_1 - x_2 = 30 - 60 = -30$$

$$P_2 = \Delta x = x_2 - x_1 = 60 - 30 = 20$$

$$P_3 = \Delta y = y_1 - y_2 = 60 - 25 = 35$$

$$P_4 = \Delta y = y_2 - y_1 = 25 - 60 = -35$$

$$Q_1 = x_1 - x_{wmin} = 30 - 20 = 20$$

$$Q_2 = x_{wmax} - x_1 = 50 - 30 = 20$$

$$Q_3 = y_2 - y_{wmin} = 60 - 20 = 50$$

$$Q_4 = y_{wmax} - y_2 = 50 - 60 = -10$$

Again,

$$U_1 = \max(0, Q_k | P_k)$$

$$U_2 = \min(0, Q_k | P_k)$$

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for u_2 ,

$$\begin{aligned}u_2 &= \max(0, q_2/p_2, q_3/p_3) \\&= \max\left(0, \frac{20}{30}, \frac{50}{35}\right) \\&= \max\end{aligned}$$

for u_2 ,

$$\begin{aligned}u_2 &= \min(0, q_2/p_2, q_4/p_4) \\&= \min\left(0, \frac{20}{30}, \frac{-20}{-35}\right) \\&= \min\left(\frac{2}{3}, \frac{2}{7}\right) \\&= \frac{2}{7}\end{aligned}$$

now,

The end points of clipped line are

$$\begin{aligned}x &= x_1 + u \Delta x \\&= 30 +\end{aligned}$$

$$\begin{aligned}y &= y_1 + u \Delta y \\&= 60 + \frac{2}{7} \times 35\end{aligned}$$

$$= 38.57$$

$$= 50$$

$$\begin{aligned}x' &= x_1 + u \Delta x \\&= 30 + \frac{2}{3} \times 30\end{aligned}$$

$$= 50$$

$$\begin{aligned}y' &= y_1 + u \Delta y \\&= 30 + \frac{2}{3} \times 35\end{aligned}$$

$$= 36.66$$

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11) Cohen - Sutherland Algorithm

Soln:

$$x_{wmin} = 50, x_{wmax} = 80$$

$$y_{wmin} = 10, y_{wmax} = 40$$

the region code for $P_1(70, 20)$

$$x - x_{wmin} = 70 - 50 = 0$$

$$x_{wmax} - x = 80 - 70 = 0$$

$$y - y_{wmin} = 20 - 10 = 0$$

$$y_{wmax} - y = 40 - 20 = 0$$

the region code for $P_2(100, 20)$

$$x - x_{wmin} = 100 - 50 = 0$$

$$x_{wmax} - x = 80 - 100 = -1$$

$$y - y_{wmin} = 10 - 10 = 0$$

$$y_{wmax} - y = 40 - 10 = 0$$

$$\text{Slope}(m) = \frac{y_2 - y_1}{x_2 - x_1} = \frac{20 - 20}{100 - 70} = \frac{-1}{3}$$

$$P_2(100, 20)$$

$$x = 80$$

$$\begin{aligned} y &= y_2 + m(x - x_2) \\ &= 20 + \frac{-1}{3}(80 - 100) \\ &= 16.67 \end{aligned}$$

$\therefore$ The new point P_2 is $(80, 16.67)$

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12) Given $P = (3, 2, 1)$

Translation matrix $T = \begin{bmatrix} 1 & 0 & 0 & -2 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 3 \\ 2 \\ 1 \\ 1 \end{bmatrix}$

$$P_1 = \begin{bmatrix} 1 \\ 0 \\ -1 \\ 1 \end{bmatrix}$$

Rotation about $60^\circ = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos 60^\circ & -\sin 60^\circ & 0 \\ 0 & \sin 60^\circ & \cos 60^\circ & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 1 \\ 0 \\ -1 \\ 1 \end{bmatrix}$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1/2 & -\sqrt{3}/2 & 0 \\ 0 & \sqrt{3}/2 & 1/2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 1 \\ 0 \\ -1 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = \begin{bmatrix} 1 \\ \sqrt{3}/2 \\ -1/2 \\ 1 \end{bmatrix}$$

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Long Question Answer

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- 1) The DDA algorithm is used to draw a straight line between two points by calculating intermediate points using incremental steps.

Working principle of DDA.

- It works by calculating the slope (m) of the line.
- The line is drawn by incrementing x & y .
- At each step, the next iteration is calculated and plotted.

Algorithm:

- 1) Start
- 2) Input end points (x_1, y_1) and (x_2, y_2)
- 3) Calculate diff
 $dx = x_2 - x_1$, $dy = y_2 - y_1$
- 4) Find num of steps $\max(dx/dy)$
- 5) Calculate increments:
 $x_{inc} = dx / \text{steps}$, $y_{inc} = dy / \text{steps}$
- 6) Initialize starting point (x, y) as (x_1, y_1)
- 7) Plot the first point (x, y)
- 8) Repeat for each step,
for $i = 1$ to step
 $x = x + x_{inc}$, $y = y + y_{inc}$
plot round (x) , round (y)
- 9) Stop

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Soln:

$$\Delta y = 3, \Delta x = 8$$

$$x_1 = 2, y_1 = 2$$

$$(x_2, y_2) = (10, 5)$$

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{5 - 2}{10 - 2} = \frac{+3}{+8} = \frac{3}{8} < 1$$

Since,

Slope (m) < 1 so it lies in x-direction

$$P_k > 0, P_{k+1} = P_k + 2\Delta y - 2\Delta x, (x_{k+1}, y_{k+1})$$

$$P_k < 0, P_{k+1} = P_k + 2\Delta y, (x_{k+1}, y_k)$$

$$\begin{aligned} \text{decision parameter } P_0 &= 2\Delta y - \Delta x \\ &= 2 \times 3 - 8 \\ &= -2 \end{aligned}$$

Iteration table

K	P_k	(x_{k+1}, y_{k+1})
0	-2	(3, 2)
1	4	(4, 3)
2	-6	(5, 3)
3	0	(6, 4)
4	-10	(7, 4)
5	-4	(8, 4)
6	2	(9, 5)
7	-8	(10, 5)

miY + Y = Y

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2) Derivation of formula:

let window : (x_w, y_w) with boundaries $x_{wmin}, x_{wmax}, y_{wmin}, y_{wmax}$

Viewport : (x_v, y_v) with boundaries $x_{vmin}, x_{vmax}, y_{vmin}, y_{vmax}$

Scaling factor:

$$S_x = \frac{x_{vmax} - x_{vmin}}{x_{wmax} - x_{wmin}}$$

$$S_y = \frac{y_{vmax} - y_{vmin}}{y_{wmax} - y_{wmin}}$$

Transformation formula:

$$T_x = \frac{(x_{wmax} - x_{vmin}) - (x_{vmax} - x_{vmin})}{x_{wmax} - x_{wmin}}$$

$$T_y = \frac{(y_{wmax} - y_{vmin}) - (y_{vmax} - y_{vmin})}{y_{wmax} - y_{wmin}}$$

Window - viewport map

$$x_v = S_x x_w + t_x$$

$$y_v = S_y y_w + t_y$$

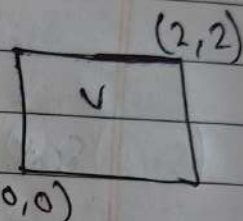
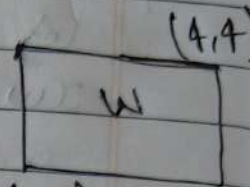
Numerical:

$$x_{wmin} = 0, \quad x_{wmax} = 4$$

$$y_{wmin} = 0, \quad y_{wmax} = 4$$

$$x_{vmin} = 0, \quad x_{vmax} = 2$$

$$y_{vmin} = 0, \quad y_{vmax} = 2$$



Scaling factors:

$$S_x = \frac{x_{vmax} - x_{vmin}}{x_{wmax} - x_{wmin}} \\ = \frac{2 - 0}{4 - 0}$$

$$= 0.5$$

$$S_y = \frac{2 - 0}{4 - 0}$$

$$= 0.5$$

Transformation formula,

$$tx = \frac{(4 - 0) - (2 - 0)}{4 - 0} \\ = 0.5$$

$$ty = \frac{(4 - 0) - (2 - 0)}{4 - 0} \\ = 0.5$$

Window viewport map (2,4)

$$x_v = 0.5 \times 2 + 0.5 = 1.5$$

$$y_v = 0.5 \times 4 + 0.5 = 2.5$$

$$(x_v, y_v) = (1.5, 2.5)$$

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Long Question

- 3) To maintain the same relative placement, the relative position in window = the relative position in viewport

$$\frac{x_w - x_{wmin}}{x_{wmax} - x_{wmin}} = \frac{x_v - x_{vmin}}{x_{vmax} - x_{vmin}}$$

$$\frac{y_w - y_{wmin}}{y_{wmax} - y_{wmin}} = \frac{y_v - y_{vmin}}{y_{vmax} - y_{vmin}}$$

3) To maintain the same relative placement, ~~the~~ the relative position in window = the relative position in viewport

$$\frac{x_w - x_{wmin}}{x_{wmax} - x_{wmin}} = \frac{x_v - x_{vmin}}{x_{vmax} - x_{vmin}}$$

$$\frac{y_w - y_{wmin}}{y_{wmax} - y_{wmin}} = \frac{y_v - y_{vmin}}{y_{vmax} - y_{vmin}}$$

4) The need of homogenous coordinate system is because:

- i) They allow all transformations (translation, rotation, scaling) to be represented using matrix multiplication.
- ii) Enable combination of multiple transformation into a single matrix.
- iii) ~~Simplify~~ Simplify computer implementation
- iv) Support 3D transformation easily.

Numerical:

Solⁿ:

$$\text{Center} = (4, 2) = (x_c, y_c)$$

$$A(0, 0), B(0, 4), C(8, 4), D(8, 0)$$

$$\text{Scaling factor (s)} = \frac{1}{2} = 0.5$$

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To scale an object relative to a fixed point (x_c, y_c) we have formula,

$$x' = x_c + s(x - x_c)$$

$$y' = y_c + s(y - y_c)$$

for $A(0,0)$

$$\begin{aligned} x' &= 4 + 0.5(0-4) \\ &= 2 \end{aligned}$$

$$\begin{aligned} y' &= 2 + 0.5(0-2) \\ &= 1 \end{aligned}$$

for $B(0,4)$

$$x' = 4 + 0.5(0-4)$$

$$= 2$$

$$y' = 2 + 0.5(4-2)$$

$$= 3$$

for $C(8,4)$

$$\begin{aligned} x' &= 4 + 0.5(8-4) \\ &= 6 \end{aligned}$$

$$\begin{aligned} y' &= 2 + 0.5(4-2) \\ &= 3 \end{aligned}$$

for $D(8,0)$

$$\begin{aligned} x' &= 4 + 0.5(8-4) \\ &= 6 \end{aligned}$$

$$\begin{aligned} y' &= 2 + 0.5(0-2) \\ &= 1 \end{aligned}$$

$\therefore$ The new coordinates of rectangle are

$$A'(2,1), B'(2,3), C'(6,3), D'(6,1)$$

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